Šūtum (2024)

Technical Rider

Goldstücke, Gelsenkirchen. Photo: Lars Gonikman

Šūtum

Short Description

Šūtum is a mixed media installation that reflects on the impermanence of laws, norms, and their interpretations. Utilizing an ultraviolet laser and photochromic pigments, the artwork creates ephemeral images of ancient laws on a stone surface that gradually disappear, symbolizing the fragility and mutability of human constructs.

The project utilizes a combination of technologies and materials. A self-assembled galvanometer scanner with a focused 405 nm laser produces both immediate and gradually evolving visual forms, while the photochromic surface allows the projected traces to emerge and fade over time.

Core Components

- Laser projection device with integrated galvo and programmed microcontroller
- 405 nm low-power laser module
- Flat stone coated with photochromic foil
- Underlying treated paper or fabric
- Focused lighting system with two narrow-angle LED spots
- Support tripods, adapters, and connection cables
- Control computer and monitor in the currently documented version

Dimensions / Duration / Scalability

- Installation footprint: 250 x 150 x 170 cm
- Packed dimensions:
- 60 x 40 x 40 cm Eurobox, estimated mass 20 kg
- 50 x 50 x 15 cm cardboard box with stone, estimated mass 12 kg
- One 120 cm roll of underlying treated paper
- Projection duration is variable
- Typical choreography runs in cycles of 4 to 5 minutes, total 3 cycles
- Project scaling is adaptable upon consultation

Spatial Requirements

- Minimum space required: 2.5 x 4 m
- A completely dark room is mandatory
- Ambient light from external sources should be reduced as much as possible
- Stable, clean, and non-reflective flooring is essential
- The space should be free from high humidity and dirt
- In the original exhibition concept, the room atmosphere may evoke a found artifact, excavation site, or abandoned building
- Surface preparation is carried out by the artist
- The underlying paper or fabric is coated on site
- Recommendations for alternative materials can be discussed in advance

Technical Requirements Provided by Venue

- 220 V / 50 Hz / max 3 A, Schuko plug system
- Dark-colored power extension with at least 5 plugs
- Power is needed for light sources, laser device, control computer, and service monitor
- Maximum distance of plugs from artwork: 4 m

- Box or masking structure for hiding the PC near the artwork
- Laser Safety Officer review is recommended for installation
- Separate power extension with visible on/off indication for the laser device as a safety measure
- Luminescent tape to limit visitor access to the stone area

Components Provided by Artist

- Laser projection device utilizing a 405 nm 20 mW laser module
- Integrated galvo and microcontroller
- Tripod for laser and one light source
- Small tripod for second light source
- Charging cables, adapters, and USB extension
- M-ITX PC with Windows and TouchDesigner in the documented version
- Portable monitor, mouse, and keyboard
- Flat stone coated with photochromic foil
- Roll of paper or fabric placed beneath the stone

Sound

The artwork does not include sound and can be installed relatively close to works that use sound, provided that the neighboring sound level is not too high.

Light

- Two adjustable LED spots with narrow focus are required
- Their positions should avoid casting shadows onto the laser path
- Reflections should be minimized

Safety

- The 405 nm laser module in the documented rider version uses 20 mW and is classified as 3B at the lower range
- Controlled conditions are required to prevent accidental exposure
- LSO assistance is recommended
- Signage and restricted access are recommended during setup
- The controlling PC should remain hidden and inaccessible to visitors
- The presence of a mediator or supervising staff member on site is highly recommended

Transport

- The work is delivered in one standard plastic Eurobox together with two additional boxes
- The artwork is fragile and must be handled with care
- Transportation safety measures are provided by the artist

Setup / Deinstallation

- Installation time: approximately 12 hours
- The artist provides on-site installation
- Assistance may be required for darkening the space and enclosing doorways with Molton
- Light positioning, laser safety adjustment, and final calibration should be carried out on site

Daily Operation

- Switch on the light sources
- Switch on the laser module
- Turn on the control computer by pressing the power button

- The system should be visually checked during operation

Maintenance

- In the documented version, the PC turns off automatically at the predefined time
- Light sources and laser module then need to be switched off manually
- Regular operational checks of the laser output, alignment, and treated surfaces are recommended
- The current exhibition version may be updated in the future, so this rider should be revised when the autonomous setup is finalized

Insurance Value

- UV laser and components: EUR 3000
- Coated surfaces: EUR 1000
- Packaging: EUR 200
- Total insurance value: EUR 4200